

Requirements Traceability Matrix (RTM)

School Management System Project

The Requirements Traceability Matrix (RTM) links business objectives and identified gaps to functional requirements, user stories, test scenarios, and success metrics. It demonstrates end-to-end traceability and shows that every requirement supports a business need. The Requirements Traceability Matrix ensures that every requirement in the School Management System is directly linked to a business objective, validated through testing, and measured against defined success criteria

RTM ID	Business Objective	Gap Description	Requirement ID	Functional Requirement	Epic / Feature	User Story ID	Priority (MoSCoW)	Test Case ID	KPI / Success Metric	Status
RTM-001	Digitize student registration	Student records are maintained manually, causing duplicate and incomplete records	FR-001	The system shall allow Admin to register a new student and generate a unique student ID	Student Registration Management / Student Registration	US-001	Must Have	TS-001	Reduce registration time by 60%	To Do
RTM-002	Assign students to classes	Class assignments are tracked manually and are difficult to update	FR-002	The system shall allow Admin to assign a registered student to a class for an academic session	Student Registration / Class Assignment	US-002	Must Have	TS-002	100% of students assigned to classes electronically	To Do
RTM-003	Digitize attendance tracking	Attendance is recorded in paper registers and reports are delayed	FR-003	The system shall allow Teachers to record daily attendance by class and date	Attendance Management/ Daily Attendance Recording	US-003	Must Have	TS-003	Reduce attendance recording time by 50%	To Do
RTM-004	Prevent duplicate attendance	Teachers may submit attendance more than once for the same class and date	FR-004	The system shall prevent duplicate attendance submissions	Attendance Management / Duplicate Attendance Prevention	US-004	Must Have	TS-004	0 duplicate attendance records	To Do
RTM-005	Notify parents of absences	Parents are informed late when students are absent	FR-005	The system shall send absence notifications to parents	Attendance Management / Parent Absence Notification	US-005	Should Have	TS-005	95% notification delivery rate	To Do
RTM-006	Record fee payments electronically	Payments are recorded manually, leading to reconciliation issues	FR-006	The system shall allow Finance Officers to record student fee payments	Fee Management / Payment Recording	US-006	Must Have	TS-006	Reduce payment posting time by 50%	To Do
RTM-007	Verify bank transfers	Unverified transfers may be posted incorrectly	FR-007	The system shall require verification before posting bank transfer payments	Fee Management / Payment Verification	US-007	Must Have	TS-007	100% of transfers verified before posting	To Do
RTM-008	Generate payment receipts	Receipts are written manually and can be lost	FR-008	The system shall automatically generate a receipt after successful payment	Fee Management / Receipt Generation	US-008	Must Have	TS-008	100% of payments produce receipts	To Do
RTM-009	Show outstanding balances	Parents have limited visibility into unpaid fees	FR-009	The system shall display fee statements and outstanding balances	Fee Management / View Outstanding Balance	US-009	Should Have	TS-009	Reduce balance enquiries by 40%	To Do
RTM-010	Capture student scores	Scores are entered in spreadsheets and are prone to errors	FR-010	The system shall allow Teachers to enter assessment and exam scores	Result Management / Enter Scores	US-010	Must Have	TS-010	Reduce score entry errors by 80%	To Do
RTM-011	Automate grade calculations	Totals and grades are calculated manually	FR-011	The system shall calculate totals, averages, and grades automatically	Result Management / Calculate Grades	US-011	Must Have	TS-011	100% automated grade calculation	To Do
RTM-012	Approve results before release	Results may be published without management review	FR-012	The system shall require Admin approval before publication	Result Management / Approve Results	US-012	Must Have	TS-012	100% of results approved before publication	To Do
RTM-013	Publish results online	Parents wait to collect printed report cards	FR-013	The system shall allow parents to view approved results online	Result Management / View Results	US-013	Should Have	TS-013	90% of results accessed online	To Do
RTM-014	Automate communications	Notifications are sent manually or inconsistently	FR-014	The system shall send automated notifications for attendance, fees, and results	Communication / Send Notification	US-014	Should Have	TS-014	95% successful message delivery	To Do
RTM-015	Provide operational dashboards	Management lacks real-time visibility into key metrics	FR-015	The system shall provide dashboards for attendance, fees, and academic performance	Reporting / View Dashboard	US-015	Could Have	TS-015	Reports available in under 5 seconds	To Do
RTM-016	Enforce access control	Sensitive information is accessible without formal controls	FR-016	The system shall enforce role-based access control	Security / Role-Based Access Control	US-016	Must Have	TS-016	100% compliance with access rules	To Do